

AMD Developer Hackathon: ACT II - Project Submission

Project: Vdcap.AI (4-Style Video Captioning Platform)

Submission Title (min 5, max 50 characters):

Vdcap.AI: Multi-Style Video Captioning Platform

Short Description (min 50, max 255 characters):

An AI-powered video captioning platform that generates 4 distinct caption styles (Formal, Sarcastic, Tech-Humor, Casual-Humor) and burns them using FFmpeg, displayed in a synchronized 4-quadrant player with rich export options.

Long Description (min 100 words, 600-2000 characters):

Vdcap.AI is a full-stack, automated video captioning platform that generates and compares four distinct caption styles side-by-side. Designed for creators and marketers, the system accepts video uploads and extracts audio and keyframes using FFmpeg and OpenCV. If audio is present, it uses cloud Speech-to-Text APIs (Groq, Sarvam, or Fireworks Whisper) for high-fidelity transcription. If the video is mute, the pipeline falls back to visual frame analysis.

The transcript and visual context are processed via Fireworks AI using the Llama-3.1-8B-Instruct and GLM models. These generate four tailored caption styles: Formal (professional), Sarcastic (witty), Humorous-Tech (developer-oriented), and Humorous-NonTech (casual). FFmpeg overlay filters burn these captions directly onto separate video streams.

On the frontend, a handcrafted dark-mode glassmorphism interface presents the results in a synchronized 4-quadrant player. Seeking, playing, or pausing any video instantly syncs the other three, allowing real-time comparison of subtitle styles. Users can export their captions as individual SRT files, structured JSON data, captioned MP4s, or a complete ZIP package including an interactive HTML summary report. Vdcap.AI also contains a modular pipeline using LoRA adapters for fine-tuning caption models on MSR-VTT and ActivityNet datasets, making it fully ready for custom AI deployment.

Event Tracks / Categories:

Track: Video-Captioning Pipeline

Category: AI-Powered Video Processing & Captioning / Multi-Model AI Pipelines

Technologies Used:

- Backend Framework: FastAPI, Python 3.10+, Uvicorn

- AI
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